

Course Code: CCE-313

ID: 2102066 ; Reg: 10193

Date: 29/07/25

P-22 } For calculating the minimum distribution time for client-server distribution, we use the following formula:

$$D_{cs} = \max \{ NF/U_s, F/d_{\min} \}$$

Similarly for calculating the minimum distribution time for P2P distribution, we use the following formula:

$$D_{p2p} = \max \{ F/U_s, F/d_{\min}, NF/(U_s + \sum_{i=1}^N U_i) \}$$

Where, $F = 15 \text{ Gbits} = 15 \times 1024 \text{ Mbps}$
 $= 15360 \text{ Mbps}$

$$U_s = 30 \text{ Mbps}$$

$$d_{\min} = d_i = 2 \text{ Mbps}$$

Calculation for Client-Server

$$\frac{F}{d_{\min}} = \frac{15360}{2} = 7680 \text{ seconds}$$

And,

For, $N = 10$;

$$\frac{NF}{U_s} = \frac{10 \times 15360}{30} = 5120 \text{ second}$$

$$\therefore D_{cs} = \max(7680, 5120) = \boxed{7680}$$

For, $N = 100$

$$\frac{NF}{U_s} = \frac{100 \times 15360}{30} = 51200$$

$$\therefore D_{cs} = \max(7680, 51200) = \boxed{51200}$$

For, $N = 1000$;

$$\frac{NF}{U_s} = \frac{1000 \times 15360}{30} = 512000$$

$$D_{cs} = \max(7680, 512000) = \boxed{512000}$$

$\therefore$ Client $\rightarrow$ Server:

	N			
	10	100	1000	
300 Kbps	7680	51200	512000	
700 Kbps	7680	51200	512000	
2 Mbps	7680	51200	512000	

Now, Calculation for P2P (Peer to Peer) Distribution :

$$D_{P2P} = \max\left(\frac{F}{U_s}, \frac{F}{d_{\min}}, \frac{NF}{U_s + \sum U_i}\right)$$

$$\therefore \frac{F}{U_s} = \frac{15360}{30} = 512 \text{ sec}$$

$$\therefore \frac{F}{d_{\min}} = \frac{15360}{2} = 7680 \text{ sec}$$

For, $N = 10$ and $U = \frac{300}{1024}$;

$$\frac{NF}{U_s + \sum U_i} = \frac{10 \times 15360}{30 + 10 \times \frac{300}{1024}} = 4665$$

$$D_{p2p} = \max(512, 7680, 4665) = \boxed{7680}$$

For, $N = 100$ and $U = \frac{300}{1024}$;

$$\frac{NF}{U_s + N \cdot U} = \frac{100 \times 15360}{30 + 100 \times \frac{300}{1024}} = \cancel{15616} \ 25904$$

$$D_{p2p} = \max(512, 7680, \cancel{25904}) = \boxed{15616} \ 25904$$

For, $N = 1000$ and $U = \frac{300}{1024}$;

$$\frac{NF}{U_s + N \cdot U} = \frac{1000 \times 15360}{30 + 1000 \times \frac{300}{1024}} = 47559$$

$$D_{p2p} = \max(512, 7680, 47559) = \boxed{47559}$$

Again - $N = 10$ and $U = \frac{700}{1024} = 0.684 \text{ Mbps}$

$$= \frac{10 \times 15360}{30 + 10 \times 0.684} = 4170$$

$$D_{p2p} = \max(512, 7680, 4170) = \boxed{7680}$$

For, $N = 100$ and $U = 0.684$

$$\frac{100 \times 15360}{30 + 100 \times 0.684} = 15616$$

$$D_{p2p} = \max(512, 7680, 15616) = \boxed{15616}$$

For, $N=1000$ and $U=0.684$ Mbps

$$\frac{1000 \times 15360}{30 + 1000 \times 0.684} = 21525$$

$$\therefore D_{p2p} = \max(512, 7680, 21525) = \boxed{21525}$$

And. When, $N=10$ and $U=2$ Mbps

$$\frac{10 \times 15360}{30 + 10 \times 2} = 3072$$

$$D_{p2p} = \max(512, 7680, 3072) = \boxed{7680}$$

For, $N=100$ and $U=2$ Mbps

$$\frac{10 \times 15360}{30 + 100 \times 2} = 6687$$

$$D_{p2p} = \max(512, 7680, 6687) = \boxed{7680}$$

For, $N=1000$ and $U=2$ Mbps

$$\frac{1000 \times 15360}{30 + 1000 \times 2} = 7576$$

$$D_{p2p} = \max(512, 7680, 7576) = \boxed{7680}$$

$\therefore$ Peer to Peer:

	N	
	10	100
300 Kbps	7680	25904
700 Kbps	7680	15616
2 Mbps	7680	7680

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